

B. Linearly Anisotropic Scattering

The anisotropic scattering cross sections for the enriched U-235 research reactor and U-D₂O reactor cases are the same as the isotropic set with the addition of the anisotropic cross sections, Σ_{22s_1} , Σ_{12s_1} , and Σ_{11s_1} .

1. Two-Group Uranium Research Reactor.

Two-Group Anisotropic Macroscopic Cross Sections

Tables 52 and 53 gives the two-group, linearly anisotropic cross sections for the research reactor. **Care must be used to correctly solve benchmark problem 71 because of the negative scattering for μ near -1.**

Table 52: Fast Group Cross Sections for Linearly Anisotropic Scattering (cm⁻¹) Research Reactor (a)

Material	ν_2	Σ_{2f}	Σ_{2c}	Σ_{22s_0}	Σ_{22s_1}	Σ_{12s_0}	Σ_{12s_1}	Σ_2	χ_2
Research Reactor (a)	2.50	0.0010484	0.0010046	0.62568	0.27459	0.029227	0.0075737	0.65696	1.0

Table 53: Slow Group Cross Sections for Linearly Anisotropic Scattering (cm⁻¹) Research Reactor (a)

Material	ν_1	Σ_{1f}	Σ_{1c}	Σ_{11s_0}	Σ_{11s_1}	Σ_{21s}	Σ_1	χ_1
Research Reactor (a)	2.50	0.050632	0.025788	2.44383	0.83318	0.0	2.52025	0.0

Infinite Medium (URRa-2-1-IN)

The test set uses the two-group enriched U-235 cross-section set from Tables 52 and 53 with $k_\infty = 1.631452$ (problem 70) and the group 2 to group 1 flux ratio = 2.614706.

One-Medium Slab Critical Dimension

The critical dimensions are listed in Table 54.

Table 54: Critical Dimension, r_c , for Two-Group Linearly Anisotropic Scattering Research Reactor (a)

Problem	Identifier	Geometry	r_c (mfp)	r_c (cm)	Reference
71	URRa-2-1-SL	Slab	6.2356	9.491600	25

One-Medium Slab Scalar Neutron Fluxes

Table 55 gives the normalized scalar neutron flux for the two-group bare research reactor (a) with linearly anisotropic scattering at four spatial points.²⁵ All values are normalized with the fast group flux at the center.

Table 55: Normalized Scalar Fluxes for Two-Group Bare Research Reactor (a)

Problem	Identifier	Geometry	Energy Group	$r/r_c = 0.20$	$r/r_c = 0.50$	$r/r_c = 0.80$	$r/r_c = 1.0$
71	URRa-2-1-SL	Slab	Fast	0.963873	0.781389	0.472787	0.189578
			Slow	0.349006	0.280870	0.157376	0.0277639

2. Two-Group U-D₂O Reactor.

Two-Group Anisotropic Macroscopic Cross Sections

Tables 56 and 57 gives the two-group, linearly anisotropic cross sections for the U-D₂O system.

Table 56: Fast Energy Group Cross Sections for Linearly Anisotropic Scattering (cm⁻¹) for U-D₂O

Material	ν_2	Σ_{2f}	Σ_{2c}	Σ_{22s_0}	Σ_{22s_1}	Σ_{12s_0}	Σ_{12s_1}	Σ_2	χ_2
D ₂ O	2.50	0.002817	0.008708	0.31980	0.06694	0.004555	-0.0003972	0.33588	1.0

Table 57: Slow Energy Group Cross Sections for Linearly Anisotropic Scattering (cm⁻¹) for U-D₂O

Material	ν_1	Σ_{1f}	Σ_{1c}	Σ_{11s_0}	Σ_{11s_1}	Σ_{21s}	Σ_1	χ_1
D ₂ O	2.50	0.097	0.02518	0.42410	0.05439	0.0	0.54628	0.0

Infinite Medium (UD2O-2-1-IN)

The test set uses the two-group linearly anisotropic D₂O cross-section set from Tables 56 and 57 with $k_\infty = 1.000196$ (problem 72) and the group 2 to group 1 flux ratio = 26.823271.

One-Medium Slab Critical Dimension

The critical dimension, r_c , is listed in Table 58.

Table 58: Critical Dimension, r_c , for Two-Group Linearly Anisotropic Scattering for U-D₂O Reactor

Problem	Identifier	Geometry	r_c (mfp)	r_c (cm)	Reference
73	UD2O-2-1-SL	Slab	336.05	1000.506133	25